

NGUYỄN GIA NGHI (Nicholas Nguyen)

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INDUSTRY EXPERIENCE

Full-Stack Engineer Intern - TMA Solutions

Jun–Aug 2026
Ho Chi Minh City

- Worked with Jira tickets spanning bug fixes, product improvements, and small features across an AI-assisted project-management application built with MongoDB, Express.js, React, and Node.js.
- Modeled and queried MongoDB data through Mongoose, built API tests, self-tested changes, and actively participated in an Agile/Scrum workflow.

SOFTWARE PROJECTS

Latency Fingerprinting | Python research engine | [Repository](#)

- Developed a versioned Python package and CLI for schema validation, testbed-record ingestion, response construction, context-aware fingerprint matching, and conservative "unknown" results; the current automated suite contains 283 passing tests.

PIXELATED Studio Edition | Creator platform & self-hosted edge-cloud-gaming research testbed

- Built and released a TypeScript/React creator platform, then extended it into a self-hosted edge-cloud-gaming testbed that streams interactive workloads from a user-owned host to a lightweight browser client.
- Implemented a hosted Fastify control plane, Electron-based host orchestration, a Dockerized runtime, GStreamer capture and encoding, secure pairing and session flows, WebRTC media and input paths, and cross-layer telemetry.

[Repository](#) | [Live application](#) | [Releases](#)

PIXELATED User Edition | Browser-based WebAssembly gaming platform

- Built and deployed a React/PWA application that runs supported NES, Game Boy, and Game Boy Color titles locally in the browser through WebAssembly-based libretro cores, without relying on the Studio desktop engine or WebRTC streaming.
- Instrumented the browser runtime to export launch-time, frame-pacing, long-task, capability, memory, and error telemetry, providing a browser-execution baseline for comparison with Studio's edge-cloud-streaming path.

[Repository](#) | [Live application](#)

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, Java, Python, C/C++, SQL
Frontend	React, Vite, PWA, browser Web APIs
Backend	Node.js, Express.js, Fastify, Spring Boot, REST APIs, Socket.IO
Data	PostgreSQL, MySQL, MongoDB
Cloud & DevOps	AWS, GCP, Linux, Docker, Kubernetes, Helm, Terraform, GitHub Actions
Systems & media	WebRTC, GStreamer, telemetry instrumentation, latency analysis

EDUCATION

VNUHCM - University of Science

Bachelor of Science | Major: Information Technology
GPA: 8.34/10.00 (3.58/4.00) | 121 credits completed

Sep 2023–May 2027
Ho Chi Minh City

CERTIFICATIONS & ENGLISH

AWS Certified Solutions Architect – Associate | [Badge](#)

AWS Certified Cloud Practitioner | [Badge](#)

IELTS Academic: 8.0 overall – Listening: 9.0, Reading: 9.0, Writing: 7.5, Speaking: 7.0 (October 2nd 2025)